

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T47 · Area and Volume of Similar Shapes

With Answers

Questions with full worked solutions

■ 25 pages

FAMILY MAP

Area Volume of Similar Shapes

F01	Use volume scale factors of similar solids	3 Q 11 marks
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F02	Link length, area and volume similarity	5 Q 21 marks
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F03	Use area scale factors of similar figures	3 Q 10 marks
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26 **A** and **B** are two similar solids.

The height of solid **A** is 31 cm

The height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**

..... cm³

(Total for Question 26 is 4 marks)

Worked Solution 31

Volumes of similar solids

UNIT 2**4 marks****Scale volumes**

$$A : B = 31 : 18.6 = 5 : 3 \Rightarrow V_A : V_B = 5^3 : 3^3 = 125 : 27.$$

Use the difference

$$V_A = 125k, V_B = 27k; \quad 98k = 735 \Rightarrow k = 7.5.$$

Find A

$$V_A = 125(7.5) = 937.5 \text{ cm}^3.$$

Final answer

$$937.5 \text{ cm}^3 \text{ (or } 938 \text{ cm}^3\text{)}$$

Question 18

4MA1-1H Q23 demand

ORIGINAL QUESTION UNIT 2

5 marks

23 Solid **A** is similar to solid **B**Here is some information about solid **A** and solid **B**

	solid A	solid B
Height (cm)	3^x	
Area (cm ²)	7776	486
Volume (cm ³)	8^x	2^{x+4}

Work out the height of solid **B**
 Give your answer as a decimal.

..... cm

Worked solution 18

Similar solids: height of B

MODULAR UNIT 2

5 marks

Demand – Similar solids: height of B

Find the linear scale factor

$$\frac{A_A}{A_B} = \frac{7776}{486} = 16 \Rightarrow k = 4.$$

Use the volume ratio

$$\frac{8^x}{2^{x+4}} = k^3 = 64 \Rightarrow 2^{2x-4} = 2^6 \Rightarrow x = 5.$$

Scale the height

$$h_A = 3^5 = 243, \quad h_B = 243/4 = 60.75 \text{ cm.}$$

Final answer

60.75 cm

Question 36

4MA1-2H Q23 whole

ORIGINAL QUESTION UNIT 2

5 marks

23 Here is a frustum of a cone.

The frustum is made by removing a small cone from a similar large cone.

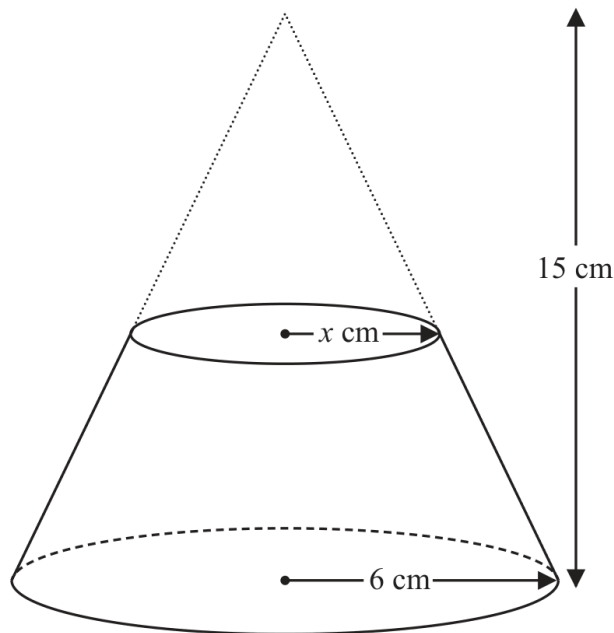


Diagram **NOT**
accurately drawn

The height of the large cone is 15 cm.

The radius of the base of the large cone is 6 cm.

The radius of the base of the small cone is x cm.

Given that the volume of the frustum is $\frac{4212}{25}\pi \text{ cm}^3$

work out the value of x

Show clear algebraic working.

$x = \dots\dots\dots$

Worked solution 36

Radius of a similar-cone frustum

MODULAR UNIT 2

5 marks

whole – Radius of a similar-cone frustum

Find the large-cone volume

$$V_L = \frac{1}{3}\pi(6^2)(15) = 180\pi.$$

Use similarity

The small cone has height $\frac{x}{6}(15) = \frac{5}{2}x$, so its volume is $\frac{1}{3}\pi x^2 \left(\frac{5}{2}x\right) = \frac{5}{6}\pi x^3$.

Use the frustum volume

$$180\pi - \frac{5}{6}\pi x^3 = \frac{4212}{25}\pi \Rightarrow \frac{5}{6}x^3 = \frac{288}{25}.$$

Solve the cubic

$$x^3 = \frac{1728}{125} = \left(\frac{12}{5}\right)^3 \Rightarrow x = \frac{12}{5} = 2.4 \text{ cm}.$$

Final answer

x = 2.4 cm

Question 18

4MA1-1H Q23

ORIGINAL QUESTION**5 marks**

- 23 A frustum is made by removing a small square-based pyramid from a similar large squared-based pyramid as shown in the diagram.

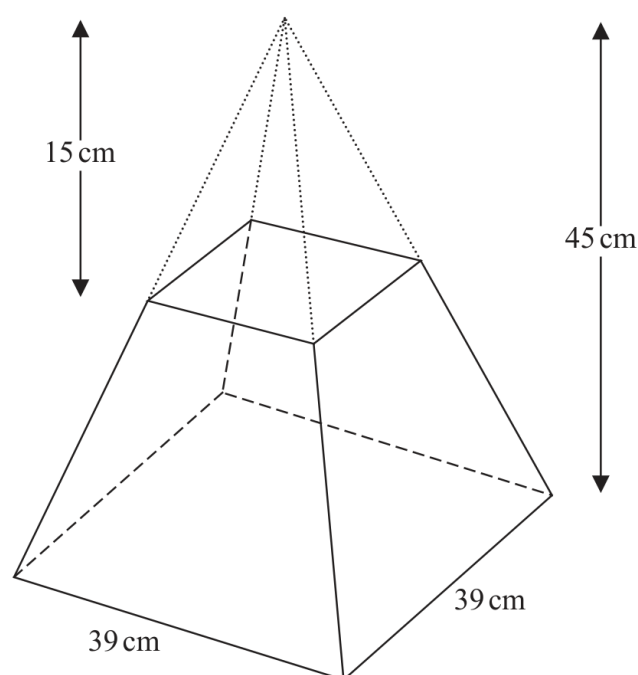


Diagram **NOT**
accurately drawn

The height of the small pyramid is 15 cm.

The height of the large pyramid is 45 cm.

The square base of the large pyramid has side length 39 cm.

Work out the **total** surface area of the frustum.

Give your answer correct to the nearest whole number.

..... cm^2

Worked Solution 18

Frustum total surface area

MODULAR UNIT 2

5 marks

Use similarity

$$\frac{15}{45} = \frac{1}{3} \Rightarrow \text{small square side} = 13.$$

Find slant heights

$$L = \sqrt{45^2 + 19.5^2} = 49.043\dots, \quad l = \frac{L}{3} = 16.348\dots$$

Find the frustum face slant

$$L - l = 32.695\dots$$

Add two squares and four trapezia

$$39^2 + 13^2 + 4 \left[\frac{1}{2}(39 + 13)(32.695\dots) \right] = 5090.3\dots$$

Final answer

$$5090 \text{ cm}^2$$

Original question 11

4MA1-1H Q17

ORIGINAL QUESTION**4 marks**

17 **A** and **B** are two similar vases.

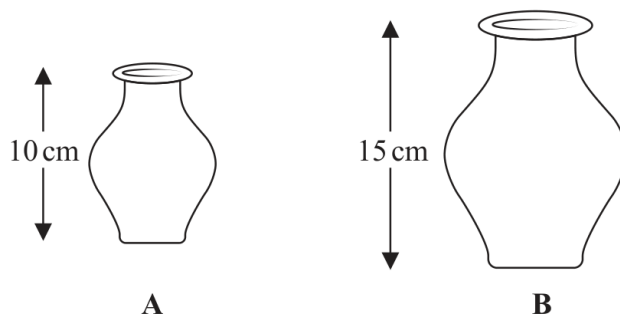


Diagram **NOT**
accurately drawn

Vase **A** has height 10 cm.

Vase **B** has height 15 cm.

The difference between the volume of vase **A** and the volume of vase **B** is 1197 cm^3

Calculate the volume of vase **A**

..... cm^3

Worked solution

Family: Use volume scale factors of similar solids • Variant: Solve a volume difference or sum for similar

METHOD / WORKING

MODULAR UNIT 2 • JANUARY 2022 DETAILED SOLUTIONS
Worked solution 11 MODULAR UNIT 2
Chapter: Area & Volume of Similar Shapes 4 marks
Q17 – Chapter: Area & Volume of Similar Shapes
Family: Use volume scale factors of similar solids • Variant: Solve a volume difference or sum for similar
solids
Method
The linear scale factor from A to B is $15/10=1.5$, so the volume scale factor is $1.5^3=27/8$.
Why: Similar solids scale volumes by the cube of the linear factor .
Method
If V is the volume of A, $(27/8)V-V=1197$.
Why: Use the stated volume difference.
Method
$(19/8)V=1197$, so $V=1197 \times 8/19$.
Why: Solve the linear equation in V .
Method
$V = 504\text{cm}^3$.
Why: Evaluate and state the volume of vase A.
Final answer
<i>volume of A : 504cm^3</i>

FINAL ANSWER

20 **A** and **B** are two similar solids.

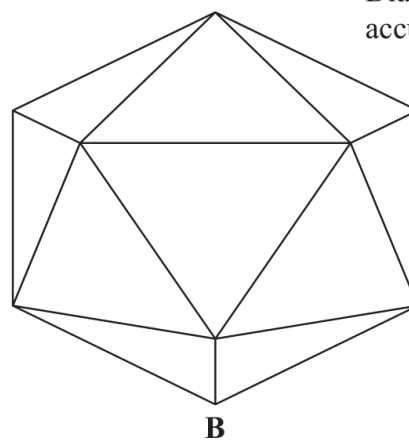
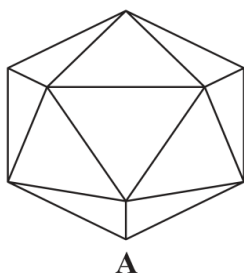


Diagram **NOT**
accurately drawn

A has a volume of 1836 cm^3

B has a volume of 4352 cm^3

B has a total surface area of 1120 cm^2

Work out the total surface area of **A**.

..... cm^2

Worked solution

January 2021 | Question 20 | 3 marks

Family: Link length, area and volume similarity | Variant: Convert between area and volume ratios

METHOD / WORKING

1. Linear scale factor $A:B = (1836/4352)^{(1/3)} = 3/4$.

2. Area scale factor is $(3/4)^2 = 9/16$.

3. $Area A = 1120 \times 9/16 = 630 \text{ cm}^2$.

FINAL ANSWER

 630 cm^2

20 Mathematically similar wooden blocks are made in a workshop.

There are small blocks and there are large blocks.

The volume of each small block is 300 cm^3

Given that

the surface area of each small block : the surface area of each large block = $25 : 36$

work out the volume of each large block.

..... cm^3

Worked solution

June 2021 | Question 20 | 3 marks

Family: Link length, area and volume similarity | Variant: Convert between area and volume ratios

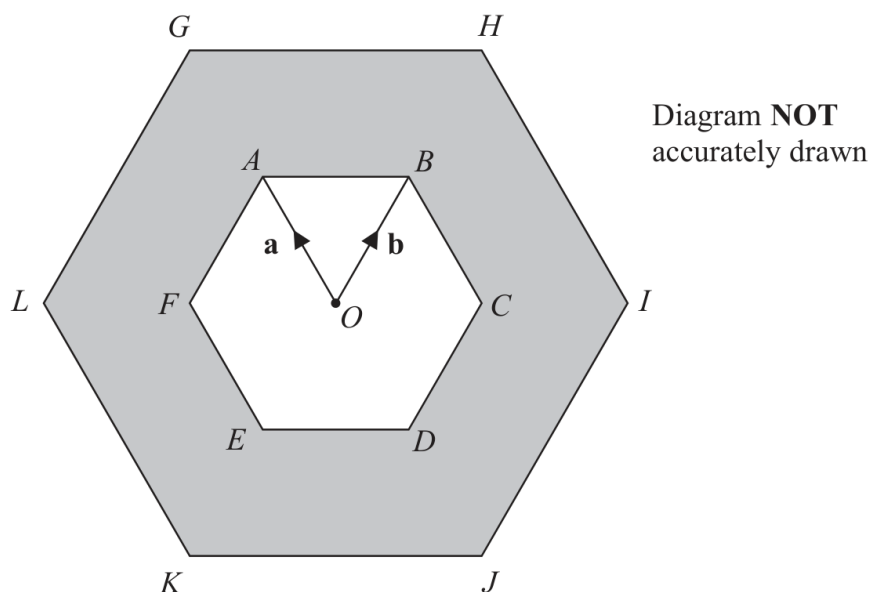
METHOD / WORKING

1. Volume ratio 36:25 gives length scale factor 6:5.
2. Large volume = $300 \times (6/5)^3$.
3. *Evaluate* to 518.4 cm³.

FINAL ANSWER

518.4 cm³

14 $ABCDEF$ and $GHIJKL$ are regular hexagons each with centre O .



$GHIJKL$ is an enlargement of $ABCDEF$, with centre O and scale factor 2

$$\vec{OA} = \mathbf{a} \quad \vec{OB} = \mathbf{b}$$

(a) Write the following vectors, in terms of $\mathbf{a}$ and $\mathbf{b}$.
Simplify your answers.

(i) $\vec{AB}$

(1)

(ii) $\vec{KI}$

(2)

(iii) $\vec{LD}$

(2)

The triangle OAB has an area of 5 cm^2

(b) Calculate the area of the shaded region.

(3)

cm^2

Worked solution

November 2021 | Question 14 | 3 marks

Family: Use area scale factors of similar figures | Variant: Find an area from a known length scale factor

METHOD / WORKING

1. (iii)

2. $\overrightarrow{LD}$ 3. $=\overrightarrow{OD}-\overrightarrow{OL}$ 4. $=-\mathbf{a}-2(\mathbf{a}-\mathbf{b})$ 5. $=2\mathbf{b}-3\mathbf{a}$ **FINAL ANSWER** $(b)90\text{cm}^2.$

20 **R** and **S** are two similar solid shapes.

Shape **R** has surface area 108 cm^2 and volume 135 cm^3

Shape **S** has surface area 300 cm^2

Work out the volume of shape **S**.

..... cm^3

(Total for Question 20 is 3 marks)

Answer reference | May-Nov 2020 | Question 20

Family: Use area scale factors of similar figures • Variant: Solve an area difference or sum for similar figures

20	$\sqrt{\frac{300}{108}}$ or $\sqrt{\frac{108}{300}}$ or $\sqrt{\frac{25}{9}}$ oe or $\sqrt{\frac{9}{25}}$ oe or $\left(\frac{300}{108}\right)^3 = \left(\frac{V}{135}\right)^2$ oe			M1 for a correct linear scale factor (fraction or ratio) or for the use of $\left(\frac{A_1}{A_2}\right)^3 = \left(\frac{V_1}{V_2}\right)^2$
	$135 \times \left(\sqrt{\frac{300}{108}}\right)^3$ oe or $\sqrt{\frac{300^3}{108^3}} \times 135^2$ or $\sqrt{390625}$			M1
		625	3	A1
Total 3 marks				

Worked solution

May-Nov 2020 | Question 20 | 3 marks

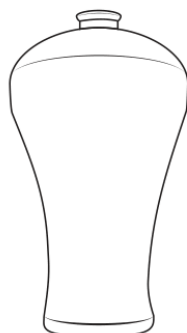
Family: Use area scale factors of similar figures | Variant: Solve an area difference or sum for similar figures

METHOD / WORKING

Worked solution
May-Nov 2020 Question 20 3 marks
Family: Use area scale factors of similar figures Variant: Solve an area difference or sum for similar figures
Method
<i>Shape R has surface area 108cm^2 and volume 135cm^3.</i>
<i>Shape S has surface area 300cm^2.</i>
Surface area ratio S:R:
$300:108=25:9$
So the linear ratio S:R is
$5:3$
Therefore the volume ratio S:R is
$5^3:3^3=125:27$
Volume of S:
$135 \times (125)/(27)=625$
Answer: <i>boxed</i> 625cm^3

FINAL ANSWER

19 **A** and **B** are two similar vases.



A



B

Diagram **NOT**
accurately drawn

The vases are such that

$$\text{surface area of vase B} = \frac{25}{64} \times \text{surface area of vase A}$$

and that

$$\text{volume of vase A} - \text{volume of vase B} = 541.8 \text{ cm}^3$$

Calculate the volume of vase **B**

..... cm^3

(Total for Question 19 is 4 marks)

Answer reference | January 2023 | Question 19

Family: Use area scale factors of similar figures • Variant: Solve an area difference or sum for similar figures

Question	working	Answer	Mark	Notes
19	$\text{eg } \sqrt{\frac{25}{64}} \left(= \frac{5}{8} = 0.625 \right) \text{ or } \sqrt{\frac{64}{25}} \left(= \frac{8}{5} = 1.6 \right) \text{ or}$ $\sqrt{25}:\sqrt{64} \text{ (5:8) or } \sqrt{64}:\sqrt{25} \text{ (8:5) or}$ $\frac{(\sqrt{25})^3}{(\sqrt{64})^3} = \left(\frac{125}{512} = 0.244140625 \right) \text{ oe or } \frac{512}{125} = 4.096$ $\frac{25^3}{64^3} = \frac{(\text{vol of B})^2}{(\text{vol of B} + 541.8)^2} \text{ or } \frac{25}{64} = \frac{(\text{vol of B})^{\frac{2}{3}}}{(\text{vol of B} + 541.8)^{\frac{2}{3}}} \text{ oe}$		4	M1 for a correct scale factor for length – may be given as a fraction or decimal or ratio or a correct scale factor for volume given as a fraction or decimal or ratio or a correct equation for the volume of vase B
	$\text{eg } \mathbf{B} \left(\frac{512}{125} - 1 \right) = 541.8 \text{ or } 3.096\mathbf{B} = 541.8 \text{ oe}$ or $\text{eg } \mathbf{A} \left(1 - \frac{125}{512} \right) = 541.8 \text{ or } \frac{387}{512} \mathbf{A} = 541.8$ $(8^3)k - (5^3)k (= 387k) = 541.8$ or $\text{eg } (k =) \frac{541.8}{387} \left(= \frac{7}{5} \right)$			M1 For a correct equation for the volume of B or a correct equation for the volume of A
	$\text{eg } (\mathbf{B}) 541.8 \div \frac{387}{125} \text{ or } 541.8 \div 3.096 \text{ or } \text{eg } 125 \times \frac{7}{5} \text{ or}$ $(\mathbf{A}) 541.8 \div \frac{387}{512} (= 716.8) \text{ oe}$			M1 For a completely correct method to find the volume of vase B or vase A
	Correct answer scores full marks (unless from obvious incorrect working)	175		A1 cao
Total 4 marks				

Worked solution

January 2023 | Question 19 | 4 marks

Family: Use area scale factors of similar figures • Variant: Solve an area difference or sum for similar figures

Official final mark-scheme pages are embedded immediately after this question.

23 Shape **P** is similar to shape **Q**

The table shows some information about shape **P** and shape **Q**

	Surface area (cm ²)	Volume (cm ³)
Shape P	200	672
Shape Q	450	

Work out the volume of shape **Q**

..... cm³

(Total for Question 23 is 3 marks)

Worked solution

November 2025 | Question 23 | 3 marks

Family: Use volume scale factors of similar solids | Variant: Solve a volume difference or sum for similar solids

METHOD / WORKING

Worked solution
November 2025 Question 23 3 marks
Family: Use volume scale factors of similar solids Variant: Solve a volume difference or sum for similar solids
Method
Surface area ratio Q:P:
$450:200=9:4$
So the linear ratio Q:P is
$3:2$
Therefore the volume ratio Q:P is
$3^3:2^3=27:8$
ShapePhasvolume 672cm^3 .
Volume of Q:
$672 \times (27)/(8)=2268$
Answer: $boxed{2268\text{cm}^3}$
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER